Calculus & Analytical Geometry – Homework 1

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Section A

Section A:

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Question No 1:-

$$M(n) = \sqrt{100 + 20n - n^2}$$

a) Domain & Range of M(n):-

Domain of M(n):-

$100 + 20n - n^2 \geq 0$

$n^2 - 20n - 100 \leq 0$

By completing square

$(n^2 - 20n + 100) - 200 \leq 0$

$(n - 10)^2 \leq 200$

$\sqrt{(n-10)^2} \leq \sqrt{200}$

Taking $\sqrt{\ }$ on both sides

$n - 10 \leq |10\sqrt{2}|$

$\boxed{n \leq 10 + |10\sqrt{2}|}$

While range of M(n) will be [redacted] $[0, 10\sqrt{2}]$

b) Intercepts:-

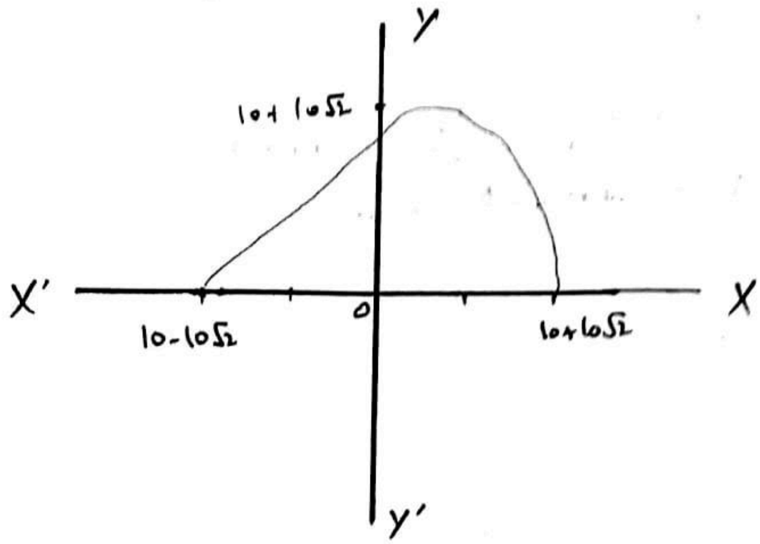
n-intercept where M is 0 will be:

$\boxed{n = 10 + |10\sqrt{2}|}$

M-intercept when n is 0 will be:-

$\boxed{M(0) = \sqrt{100 + 0 - 0} \Rightarrow 10}$

c) Sketch the graph..



M(n) will be a semicircle.

d) Practical Domain & Range as per Context

- Since 'n' will be the number of processes that consume memory, they can not be negative.
  So, domain will be: $[0, 10+|10\sqrt{2}|]$
- Hence range (memory consumption) also varies to $[2, 10\sqrt{2}]$

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Question No 2:-

$$T(n) = \frac{3n+2}{n-4}$$

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• Domain of $T(n)$:-

Mathematically, $n-4 \neq 0$ for a real output. Hence domain can be $\mathbb{R} - \{4\}$.

But, since **n** is input size, $n-4 > 0$; $n > 4$

So the domain will be $[5, \infty)$.

• Range of $T(n)$:-

Since the minimal size is '5', the range will be $[17, \infty)$.

Question No 3:-

$$f(x) = \ln|x-2| + 1$$

a) Domain & Range of $f(x)$:-

~~Since ln|x-2| ... , we~~

• Since natural log doesn't accept '0' as an input, $x-2 \neq 0$; $x \neq 2$

So, domain of $\ln|x-2|+1$ is $\mathbb{R} - \{2\}$.

• Range of $f(x)$ will be $\mathbb{R}$.

b) Intercepts of $f(x)$:-

x-intercept as $y=0$:-

$$\ln|x-2| + 1 = 0$$

$$\ln|x-2| = -1$$

~~M~~ Taking antilog on both sides..

$|x-2| = e^{-1}$

$x - 2 = |e^{-1}|$

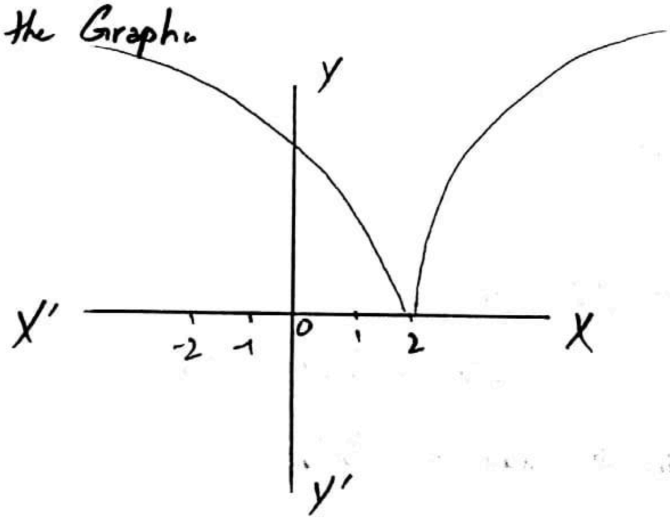
$\boxed{x = |e^{-1}| + 2}$

- **y-intercept** at $f(0)$ is..

$f(0) = \ln|-2| + 1$

$\boxed{f(0) = \ln 2 + 1}$

c) Sketch the Graph..



d) $x \neq 2$ explanation:

Natural log is only defined for positive number arguments. $\ln(0)$ or $\ln(-x)$ are undefined and hence don't give a real output.

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Question No 4:-

$$S(d) = {}^{10}/_{\sqrt{d^2+4}} \quad , d \geq 0$$

$S(d) = \dfrac{10}{\sqrt{d^2+4}} \quad \Rightarrow \quad \dfrac{10}{u(d)} \; : \; u(d) = \sqrt{d^2+4}$

$\sqrt{d^2+4} \neq 0 \quad , d \geq 0$

$(\sqrt{d^2+4})^2 \geq (0)^2$

$d^2+4 \geq 0$

$d^2 \geq -4$

$\sqrt{d^2} \geq \sqrt{-4} \quad \therefore$ No real 'd'.

- However, for any value of 'd', the radical $\sqrt{d^2+4} \geq 0$

Since 'd' is distance, so the domain can only be positive numbers or '0'. So, **Domain = $[0, \infty)$**

- Since 'd' is in denominator, so higher the value of 'd', smaller the range. So the range at max will be at least 'd'.

$S(d)$ at 0: $S(0) = \dfrac{10}{\sqrt{4}} \Rightarrow 5$

So: **Range = $(0, 5]$**

Question No 5.

$$E(r) = |r^2 - 4r + 3|, \quad 0 \leq r \leq 4$$

a) Domain & Range of $E(r)$:-

$$|r^2 - 3r + r + 3| = |(r-3)(r-1)|$$

As per the given restriction, domain is defined at all values so, **Domain = [0, 4].**

- Since it's a parabola facing upward, it's negative b/w roots. Means (1, 3) is negative, while rest is positive. Now for all values, range is : **[0, 3].**

b) Intercepts :-

- r-intercept = $r = 1$ & $r = 3$; $(1, 0)$ & $(3, 0)$
- $E(r)$-intercept = $E(0) = 0 - 0 + 3 \Rightarrow (0, 3)$

c) $r^2 - 4r + 3$ is +ve / -ve :-

- Since roots are (1 & 3), ~~so~~ and domain is from [0, 4].
- Now the interval is $[0,1) \cup (1,3) \cup (3,4]$
- From (1, 3) the sign is negative, else positive.

So $r^2 - 4r + 3 > 0$ at $[0, 1) \cup (3, 4]$

$r^2 - 4r + 3 < 0$ at $(1, 3)$.

$r^2 - 4r + 3 = 0$ at $\{1, 3\}$.

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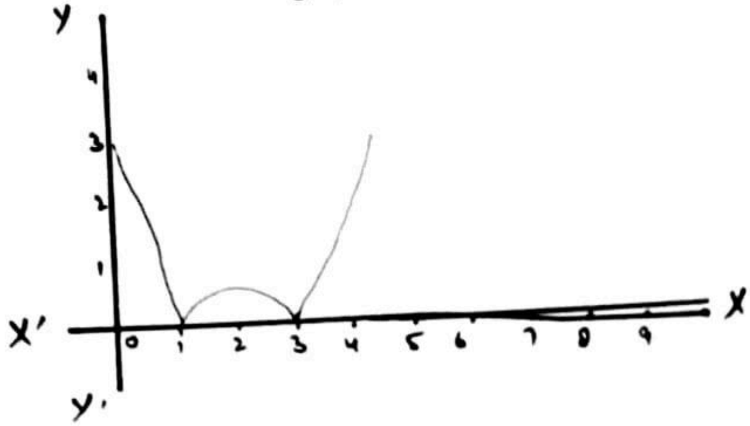
84 d) Sketch the graph.



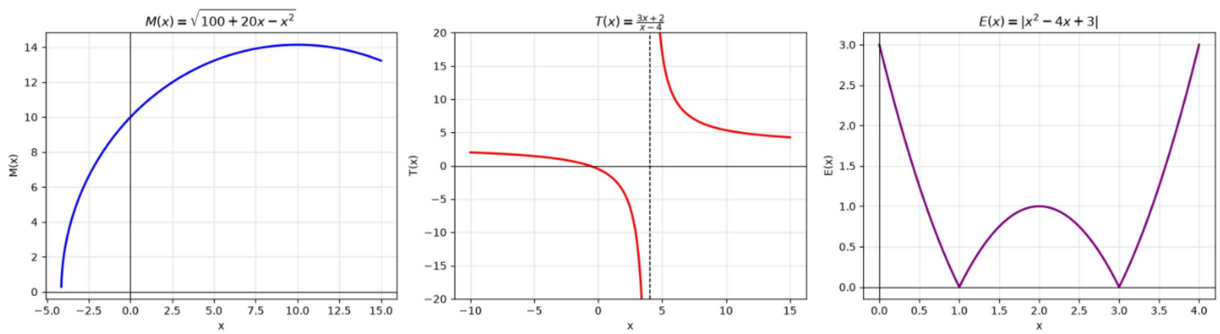
e) Role of Absolute

The value of $x^2 - 4x + 3$ was -ve from (1, 3)
Absolute function bounced it back to positive y-axis, else it would've been negative.

Section B:

Question 6

a) Screenshot



b) Domain of M(x)

The domain and range of graph 1 are the same as what I drafted **algebraically.**
Estimated Domain: [0, 10+|10*(2)^(½) |]
Range: [0, 10*(2)^(½)]

c) Domain of T(x)

The excluded value of x is **4** in the second graph because it causes an undefined behaviour in the function and hence the graph can't attain this value. Same goes with the **y = 3**.
Domain: Real Numbers - {4}
Range: Real Numbers - {3}

d) Domain of E(x)

Domain: [0,4]
Range: [0,3]

e) Intercepts of all Graphs

M(x) intercepts:
x-intercept: Both start and end point of graph where it touches the x axis (y = 0). Given this graph that was plotted, it's only at 10-10*(2)^(½)
y-intercept: (0,10)

T(x):
x-intercept: (-1.666, 0)
y-intercept: (0, -0.5)

E(x):
x-intercept: (1,0) & (3,0)
y-intercept: (0,3)

f) Advantage & Disadvantages of using Computational Graphs

- Helps to visually understand functions and detect asymptotes, discontinuity, etc.
- May not plot values perfectly.

Question 7

a) Screenshot

b) Domain & Range

Domain: [0,5]
Range: [3,23]

c) Predicted Score at x = 2.5

y = 4x + 3
y = 4*2.5 + 3
y = 13

d) Slope and Vertical Intercept meaning

- Slope means that with each study hour, quiz score increases by **4**
- Vertical intercept at **3** means the score a student may score with no study based on prior knowledge, flukes or by attending classes only.

e) Negative Study Time & Study time beyond interval

- No, the model to predict negative study sessions is illogical. Time is a quantity which can't be negative.
- Using the model for predicting quiz scores for study hours beyond 5 will work only if we have a maximum scores interval. Using without an interval is also illogical as there's a certain limit to both how much a student can study and the maximum score of the quiz